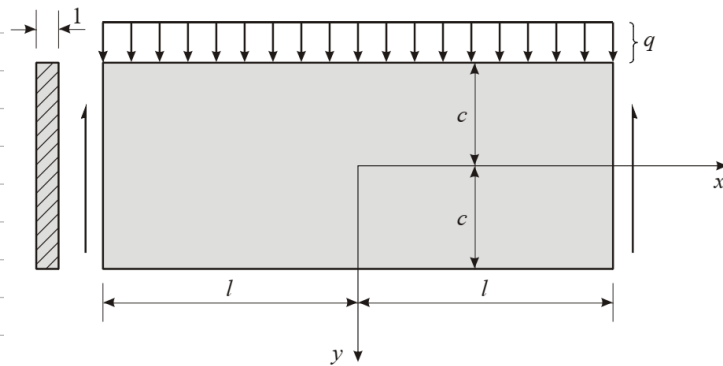




① Verificar equilibrio

② Verificar CB

- Verificar las ecuaciones de compatibilidad (sabiendo que es un material elástico lineal) → No se pide

$$\sigma_{xx} = \frac{q}{2I}(l^2 - x^2)y + \frac{q}{2I}\left(\frac{2}{3}y^3 - \frac{2}{5}c^2y\right)$$

$$\sigma_{yy} = -\frac{q}{2I}\left(\frac{1}{3}y^3 - c^2y + \frac{2}{3}c^3\right)$$

$$\sigma_{xy} = -\frac{q}{2I}(c^2 - y^2)x$$

$$\sigma_{zz} = \sigma_{xz} = \sigma_{zy} = 0$$

①

$$\frac{\partial \sigma_{ij}}{\partial x_i} + \cancel{X_j} = 0$$

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0$$

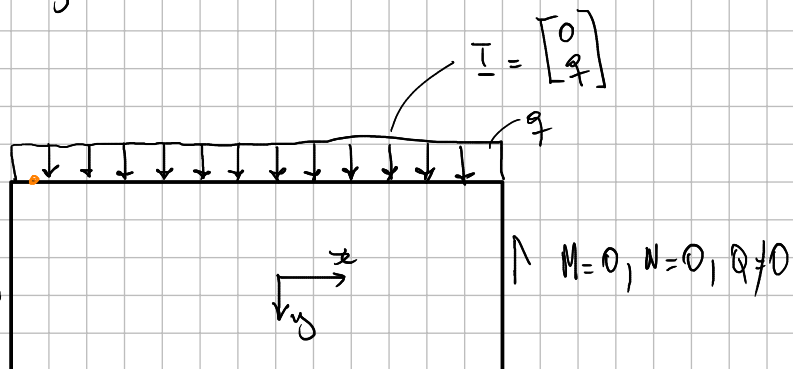
$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0$$

→ No hay fuerzas de cuerpo

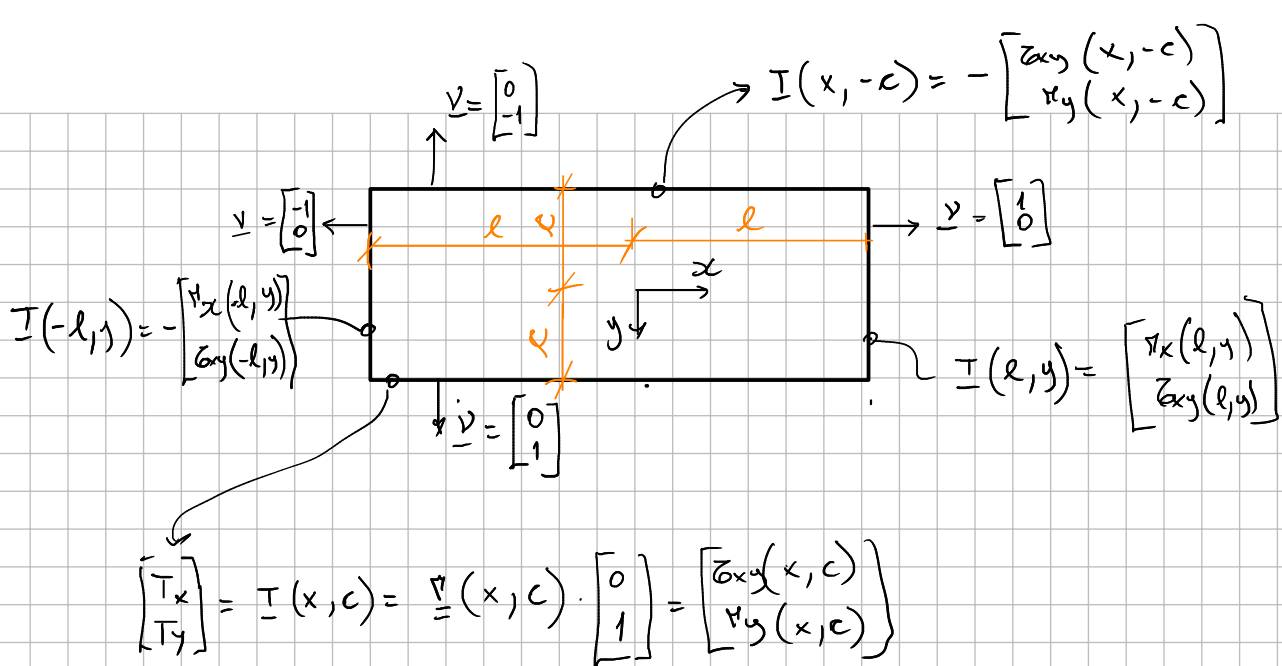
Verificar equilibrio

②

Esf. resultantes
 $M=0, N=0, Q \neq 0$
 momento normal corte



$$\begin{matrix} M=0 \\ N=0 \\ Q=0 \end{matrix} \quad \text{ó} \quad \underline{I} = \underline{0}$$



$\underline{v} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$
 $\underline{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $\underline{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 $\underline{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$
 $I(x, -c) = - \begin{bmatrix} \sigma_{xy}(x, -c) \\ \tau_{xy}(x, -c) \end{bmatrix}$
 $I(l, y) = \begin{bmatrix} \tau_{xy}(l, y) \\ \sigma_{xy}(l, y) \end{bmatrix}$
 $I(-l, y) = - \begin{bmatrix} \tau_{xy}(-l, y) \\ \sigma_{xy}(-l, y) \end{bmatrix}$
 $I(x, c) = \begin{bmatrix} \sigma_{xy}(x, c) \\ \tau_{xy}(x, c) \end{bmatrix}$
 $\underline{T} = \begin{bmatrix} T_x \\ T_y \end{bmatrix} = \underline{I}(x, c) = \underline{v}(x, c) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sigma_{xy}(x, c) \\ \tau_{xy}(x, c) \end{bmatrix}$

• Para los lados con normal $\pm y$

$$\sigma_{xy}(x, \pm c) = -\frac{q}{2I} [c^2 - (\pm c)^2] \cdot x = 0$$

$$\tau_{xy}(x, \pm c) = -\frac{q}{2I} \left(\frac{1}{3} (\pm c)^3 - c^2 (\pm c) + \frac{2}{3} c^3 \right)$$

$$\hookrightarrow I = \frac{2c^3}{3}$$

$$= -\frac{1}{4} q (\pm 1 \mp 3 + 2) \begin{cases} \tau_{xy}(x, +c) = 0 \\ \tau_{xy}(x, -c) = -q \end{cases}$$

* Lado inferior

$$\underline{I}(x, c) = \begin{bmatrix} \sigma_{xy}(x, c) \\ \tau_{xy}(x, c) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \text{Verifica los CB}$$

* Lado superior

$$\underline{I}(x, -c) = - \begin{bmatrix} \sigma_{xy}(x, -c) \\ \tau_{xy}(x, -c) \end{bmatrix} = - \begin{bmatrix} 0 \\ -q \end{bmatrix} = \begin{bmatrix} 0 \\ q \end{bmatrix} \Rightarrow \text{Verif. los CBs}$$

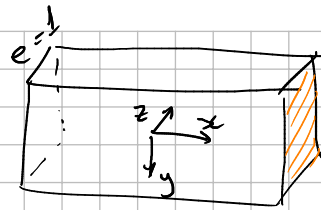
• Para los lados con normal $\pm x$:

$$\tau_{xy}(\pm l, y) = \frac{q}{2I} [l^2 - (\pm l)^2] + \frac{q}{2I} \left(\frac{2}{3} y^3 - \frac{2}{3} c^2 y \right)$$

$$\sigma_{xy}(\pm l, y) = -\frac{q}{2I} (c^2 - y^2) \pm l = \mp \frac{ql}{2I} (c^2 - y^2)$$

x lado derecho

$$\underline{T}(l, y) = \begin{bmatrix} T_x(l, y) \\ T_y(l, y) \end{bmatrix}$$



$$\begin{bmatrix} N \\ Q \end{bmatrix} = \iint_A \underline{T}(l, y) \cdot dA = \int_y \int_z \underline{T}(l, y) dz = \int_{-c}^c \underline{T}(l, y) \cdot \underbrace{\int_0^1 1 dz}_{1} dy$$

$$= \begin{bmatrix} \int_{-c}^c T_x(l, y) dy \\ \int_{-c}^c T_y(l, y) dy \end{bmatrix} = \begin{bmatrix} \int_{-c}^c T_x(l, y) \cdot dy \\ \int_{-c}^c T_y(l, y) \cdot dy \end{bmatrix}$$

$$\int_{-c}^c T_x(l, y) \cdot dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} y^3 - \frac{2}{5} c^2 y \right) \cdot dy =$$

$$= \frac{q}{2I} \left(\frac{1}{6} y^4 - \frac{1}{5} c^2 y^2 \right) \Big|_{-c}^c = 0$$

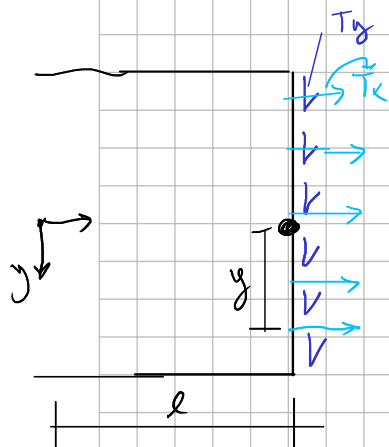
$$\int_{-c}^c T_y(l, y) \cdot dy = \int_{-c}^c -\frac{q l}{2I} (c^2 - y^2) dy = -\frac{q l}{2I} \left(c^2 y - \frac{y^3}{3} \right) \Big|_{-c}^c$$

$$= -q \cdot l$$

$$\begin{bmatrix} N \\ Q \end{bmatrix} = \begin{bmatrix} 0 \\ -q l \end{bmatrix}$$

coincide con el esfuerzo resultante de CP

signo negativo $\Rightarrow$ va hacia arriba $\Rightarrow$



$$M = \iint_A \widehat{T_x} \cdot dA \cdot y = \iint_A T_x \cdot y \cdot dA =$$

$$= \int_{-c}^c T_x(l, y) \cdot y dy = \int_{-c}^c \frac{q}{2I} \left(\frac{2}{3} y^4 - \frac{2}{5} c^2 y^2 \right) dy$$

$$= \frac{q}{2I} \left(\frac{2}{3} \frac{y^5}{5} - \frac{2}{5} c^2 \frac{y^3}{3} \right) \Big|_{-c}^c = 0 \Rightarrow \text{Verificar los CBS}$$

• ~~de~~ izquierdo

$$\tau_x(-l, y) = \tau_x(l, y)$$

$$\begin{aligned} \begin{bmatrix} N \\ Q \end{bmatrix} &= \iint_A \underline{T}(-l, y) \cdot dA = - \begin{bmatrix} \int_{-c}^c \tau_x(-l, y) \cdot dy \\ \int_{-c}^c \tau_{xy}(-l, y) dy \end{bmatrix} = \\ &= \begin{bmatrix} 0 \\ \int_{-c}^c \underbrace{\tau_{xy}(-l, y)}_{\tau_{xy}(l, y)} \cdot dy \end{bmatrix} = \begin{bmatrix} 0 \\ -ql \end{bmatrix} \Rightarrow \text{Verif. } \textcircled{B} \end{aligned}$$

$$\begin{aligned} M &= \iint_A \tau_x(-l, y) \cdot y \cdot dA = \int_{-c}^c \underbrace{\tau_x(-l, y)}_{\tau_x(l, y)} \cdot y \cdot dy = - \int_{-c}^c \underbrace{\tau_x(l, y) \cdot y}_{\textcircled{0}} \cdot dy \\ &= -0 = 0 \Rightarrow \text{Verif. con } \textcircled{B} \end{aligned}$$